Open-vocabulary detections are lifted into the global frame using depth and camera pose. Same-category observations are associated with mapped instances through spatial consistency, while unassociated detections are accumulated across views for subsequent scene-graph revision.

For tasks configured for semantic recovery, LogicFly distinguishes three levels of negative evidence. A **pose-level miss** removes the attempted candidate while keeping the task active. **Region-level exhaustion** invokes alternative room–functional-region hypotheses grounded in the hierarchy; these hypotheses become geometric observation regions and are ranked using semantic relevance, coverage, and travel cost. A **task-level negative outcome** is produced when the task-specific stopping condition is reached, after which CPTG updates downstream logic. Within semantic recovery, newly covered portions of an unproductive region reduce its belief, shifting search toward alternative hypotheses.

Novel detections are accumulated online through spatially consistent multi-view fusion and materialized in the updated scene graph after execution. Thus, task states and semantic-region beliefs can affect replanning within the current execution, whereas newly created object instances support later use of the revised map.

After each terminal verification, the system updates the task state, candidate poses, and semantic search prior, recomputes $\mathcal{E}_t$, and invokes JTVP from the current robot pose. When a task reaches a task-level outcome, CPTG additionally updates $\mathcal{R}_t$ and evaluates $\mathcal{R}_{\text{cond}}$ to activate or skip downstream conditional tasks. This closes the loop among task logic, 3D observation, geometric motion, and online semantic evidence.

G. Methodology Summary

LogicFly presents a hierarchical perception–reasoning–motion framework for long-horizon aerial navigation. It first builds a semantic–geometric representation that grounds object evidence and room structure in observed geometry. Natural-language instructions are then organized into a Conditional Partial-Order Task Graph (CPTG), enabling precedence constraints and observation-dependent branches to be reasoned about jointly with task-conditioned viewpoints. Within a receding-horizon planning scheme, the system coordinates task selection, viewpoint generation, and collision-aware motion, while safe-corridor-constrained MINCO optimization converts the selected discrete route into a dynamically feasible trajectory. Terminal observations close the loop by verifying task outcomes and updating semantic evidence, candidate viewpoints, and task states. Once a task reaches its terminal outcome, the updated semantic confidence is used to revise the semantic map by incorporating newly verified instances and the corresponding scene-graph updates. Through this integration of semantic task reasoning and geometric motion generation, LogicFly supports reliable execution of complex multi-target instructions in unknown indoor environments.

V. Experiments

We evaluate LogicFly through simulation, ablation studies, and real-world UAV experiments, covering benchmark performance, the contributions of CPTG and JTVP, and practical deployment.

A. Experimental Setup

Evaluation benchmark. We evaluate all methods on six HM3D [39] scenes: 00070, 00166, 00273, 00290, 00337, and 00409. The benchmark contains 152 language instructions, including 61 short single-target instructions, 60 long-horizon instructions confined to a single floor, and 31 long-horizon cross-floor instructions. Using the valid initial-state seeds and two repeated trials, each method is evaluated over 854 episodes. The benchmark covers ordered execution, conditional branches, target absence, failure recovery, spatial relations, and instance disambiguation.

Evaluation metrics. We report task Success Rate (SR), Success weighted by Path Length (SPL), and Completion Time (CT). An episode is considered successful only when all required task outcomes are completed while satisfying the instruction-level ordering, branching, spatial, recovery, and disambiguation constraints. Reaching the correct objects without satisfying the complete instruction is counted as a failure. For episode e, SPL is computed as

$$\text{SPL}_e = S_e \frac{L_e^{\text{ref}}}{\max(L_e^{\text{ref}}, L_e)}, \tag{15}$$

where S_e is the strict binary task-success indicator, L_e is the executed trajectory length, and L_e^{ref} is the shortest feasible path length under the same task constraints. SPL is not applicable to target-absence episodes.

Baselines. We compare LogicFly against four representative map-based and spatial-reasoning navigation methods. OneMap [22] is a zero-shot multi-object navigation method that constructs a real-time open-vocabulary feature map and queries the unified map to locate multiple target categories. MapGPT [6] is an LLM-based VLN method that incorporates an environment map into the language prompt and adaptively updates its planned route according to the explored topology. SpatialGPT [7] performs zero-shot VLN by maintaining a structured spatial memory and applying spatial chain-of-thought reasoning to select navigation actions. SpatialNav [28] represents perceived objects and their spatial relations using a spatial scene graph, which supports graph-conditioned reasoning for zero-shot navigation.

Implementation details. LogicFly first compiles each instruction into a conditional and precedence-constrained task graph, which is subsequently executed by the proposed task-graph runner. Reusable semantic maps are constructed from registered RGB-D observations. The mapping front end uses SAM 3.1 in image mode with a fixed vocabulary of 30 indoor object categories and a confidence threshold of 0.50. Video tracking is disabled. Boxer lifts the accepted masks into 3-D and performs multi-view instance fusion using an IoU threshold of 0.20, an instance-confidence threshold of 0.45, and a minimum of four observations. During task execution, online object evidence is obtained from calibrated RGB-D observations using open-vocabulary recognition and is projected into the world frame using the estimated camera pose and aligned depth.